

Algebra & Equations



Solving linear equations

- 1 Solve $3x - 7 = 11$.
 - 2 Solve $5(x - 2) = 3x + 4$.
 - 3 Solve $\frac{3x - 2}{4} = x - 1$.
 - 4 Solve $2(x + 3) - 5 = 3(x - 1)$.
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Expanding and simplifying expressions

- 5 Expand and simplify $3(2x - 1) - 2(x - 4)$.
 - 6 Expand $(x + 5)(x - 2)$.
 - 7 Expand $(2x - 3)^2$.
 - 8 Simplify $\frac{6x^2 - 9x}{3x}$ (for $x \neq 0$).
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Factoring quadratic expressions

- 9 Factor $x^2 + 7x + 12$.
 - 10 Factor $x^2 - 2x - 15$.
 - 11 Factor $2x^2 + 5x - 3$.
 - 12 Factor $x^2 - 49$ using the difference of squares.
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Solving quadratic equations by factoring

- 13 Solve $x^2 - 5x + 6 = 0$.
- 14 Solve $x^2 + 2x - 8 = 0$.
- 15 Solve $2x^2 - 5x - 3 = 0$.

16 Solve $x^2 - 9x + 20 = 0$.

The quadratic formula and the discriminant

17 Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.

18 Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.

19 Find the discriminant of $x^2 - 6x + 9 = 0$ and state how many real roots it has.

20 Use the quadratic formula to solve $3x^2 - 2x - 1 = 0$.

Absolute value equations

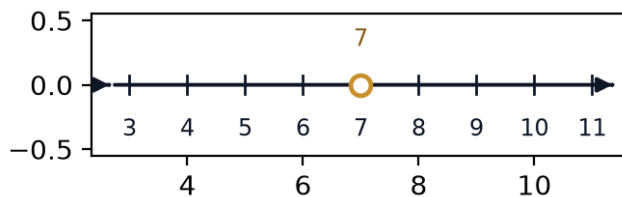
21 Solve $|x - 4| = 9$.

22 Solve $|2x + 3| = 7$.

23 Solve $3|x - 1| = 12$.

Linear inequalities

24 Solve $2x - 5 < 9$ and graph the solution set on a number line.



25 Solve $-3x + 4 \geq -8$.

26 Solve the compound inequality $5 < 2x + 1 \leq 11$.

Systems of linear equations

27 Solve the system $x + y = 10$, $x - y = 2$.

- 28 Solve the system $3x + 2y = 16$, $x - y = 2$.
- 29 This question has two parts. The sum of two numbers is 15 and their difference is 3.
- a Write a system of two equations for the numbers x and y .
 - b Solve the system to find both numbers.
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Rational equations

- 30 Solve $\frac{x}{x-2} = 3$.
- 31 Solve $\frac{2}{x+1} = \frac{1}{3}$.
- 32 Solve $\frac{x+1}{x-1} = 2$.
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Solving for a variable in a formula

- 33 The area of a triangle is $A = \frac{1}{2}bh$. Solve for h .
- 34 The circumference of a circle is $C = 2\pi r$. Solve for r .
- 35 The formula $v = v_0 + at$ gives velocity under constant acceleration. Solve for t .
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Linear-quadratic systems

- 36 Solve the system $y = x^2 - 1$, $y = 2x + 2$.
- 37 Solve the system $y = x^2$, $y = 3x - 2$.
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Word problems: setting up and solving equations

- 38 This question has two parts. A rectangular garden has a length that is 4 meters more than its width w . Its area is 96 square meters.
- a Write an equation relating w to the area.
 - b Solve for w , and state the garden's length.

- 39** This question has two parts. A mobile-data plan charges a fixed monthly fee f plus a rate r per gigabyte used. One month, 5 GB of data cost a total of 65 dollars; another month, 9 GB cost a total of 93 dollars.
- a** Write a system of two equations for f and r .
 - b** Solve the system to find the fixed fee and the rate per gigabyte.